Citi Bike and Weather Exploratory Data Analysis

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1 Introduction

1.1 Problem Statement

This EDA study examines how weather conditions affect Citi Bike usage in Jersey City. We quantify the sensitivity of cycling behavior to precipitation, wind speed, and temperature. We also analyze the differences between weekday and weekend travel patterns to see how commuting and leisure trips respond to external conditions, based on the hourly data. Finally, we investigate user preferences for electric bikes (e-bikes) during challenging weather.

1.2 Dataset Description

We use two datasets for this analysis. The main one is the Jersey City Citi Bike System Data, a subset of a public repository of Citi Bike trip records managed by Lyft. The dataset tracks individual journeys, including ride IDs, bicycle types (classic and electric), and precise timestamps for both departures and arrivals. Spatial details are captured through station names and geographic coordinates, while user status is either annual Members or Casual riders.

These mobility records are integrated with localized meteorological data from the Open-Meteo Historical Weather API, which provides hourly and daily readings for temperature, precipitation, and wind speed. By aligning both sources by date and time, the final dataset enables a direct correlation between urban cycling volume and specific atmospheric conditions. We also segmented the data into weekday and weekend groups to compare patterns driven by work schedules and recreational activities.

2 Basic Data Exploration

We first look at the overall distribution of rides. Figure 1 shows the frequency distribution of daily ride counts, and most days have 3000-4000 rides. Figure 2 indicates that weekdays have a slightly higher average ride counts than weekends, suggesting that Citi Bike is mostly used for commuting rather than just recreation. Figure 3 groups total rides by month and shows clear seasonal changes. Ridership peaks during the late spring, summer, and early autumn, then drops significantly in the winter. Figure 4 supports this by plotting daily average temperature against total rides, showing a positive linear correlation where warmer days lead to more rides.

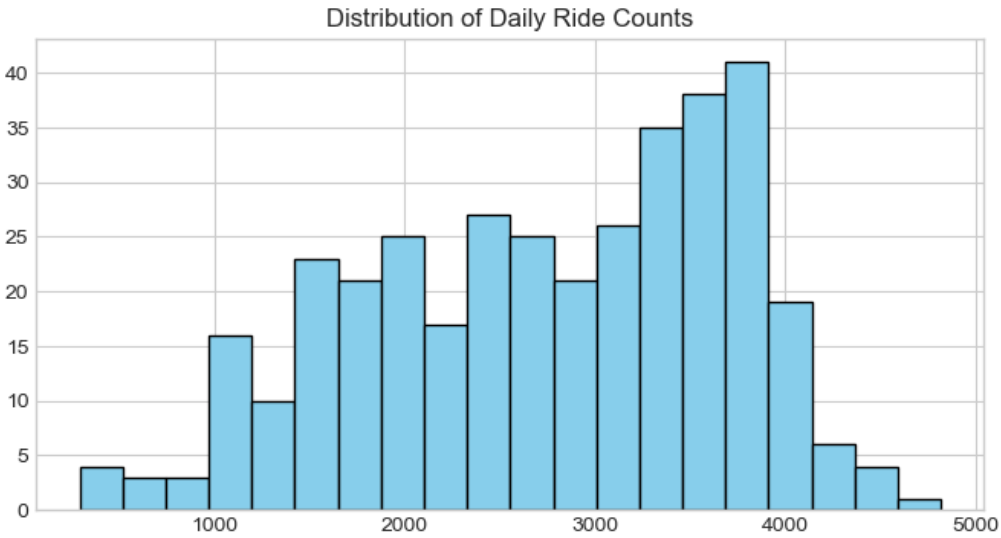

Figure 1: Daily Ride Counts Distribution

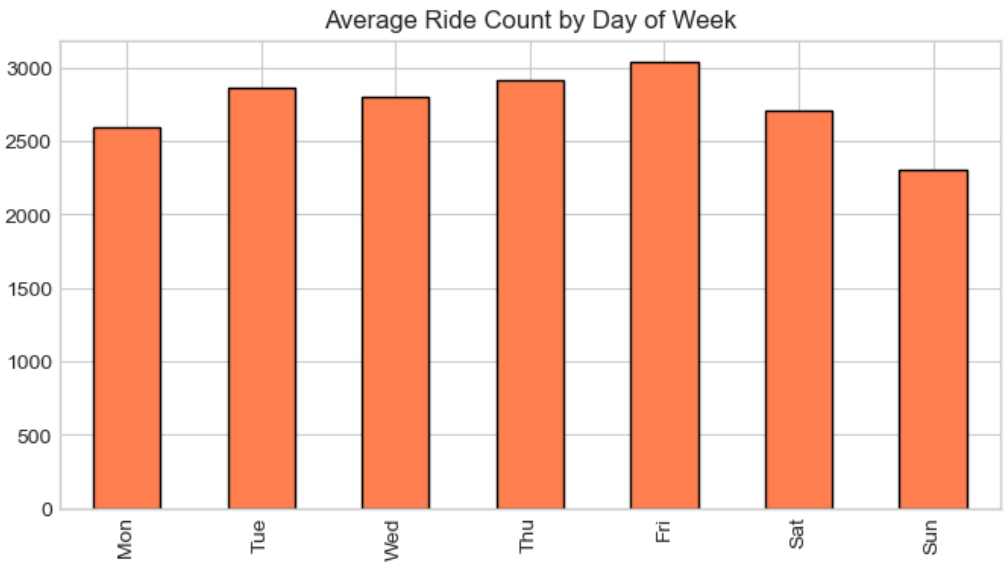

Figure 2: Average Rides by Day of Week

Figure 3: Total Rides by Month

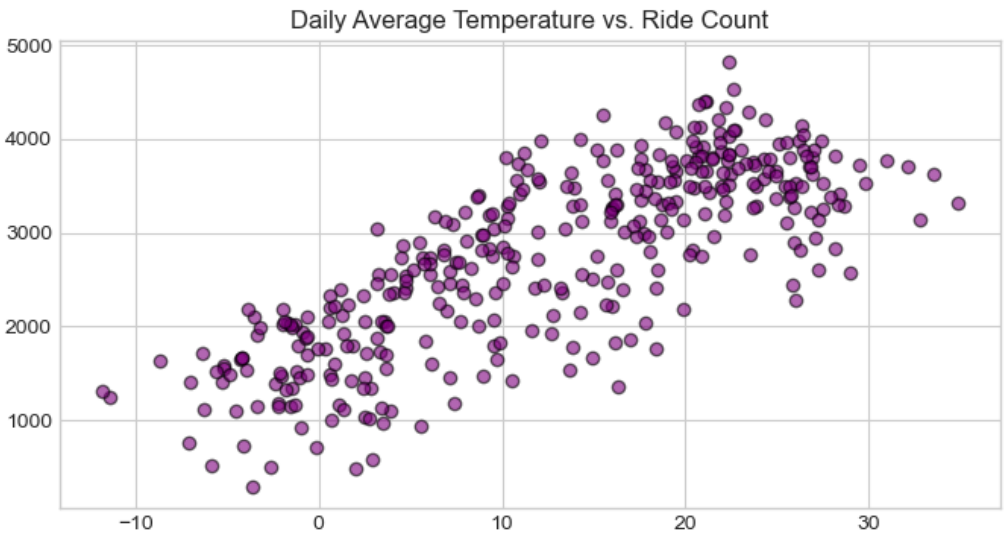

Figure 4: Temperature vs. Daily Rides

3 Overall Weather Impact

Weather conditions directly affect daily ridership. Figure 5 groups average rides by temperature ranges with a step of five degree, showing a clear increase in rides as temperatures rise. Also, there is a slight drop on the daily ride when there is high temperture (over 25 degree). On the other hand, adverse conditions reduce ridership. Figure 6 shows that any precipitation lowers ride counts. Light rain causes a small decrease, and heavy rain leads to noticeable drops in ridership. Wind speed also has a negative impact. Figure 7 shows that high wind speeds consistently reduce ride counts, as windy conditions make riding more difficult.

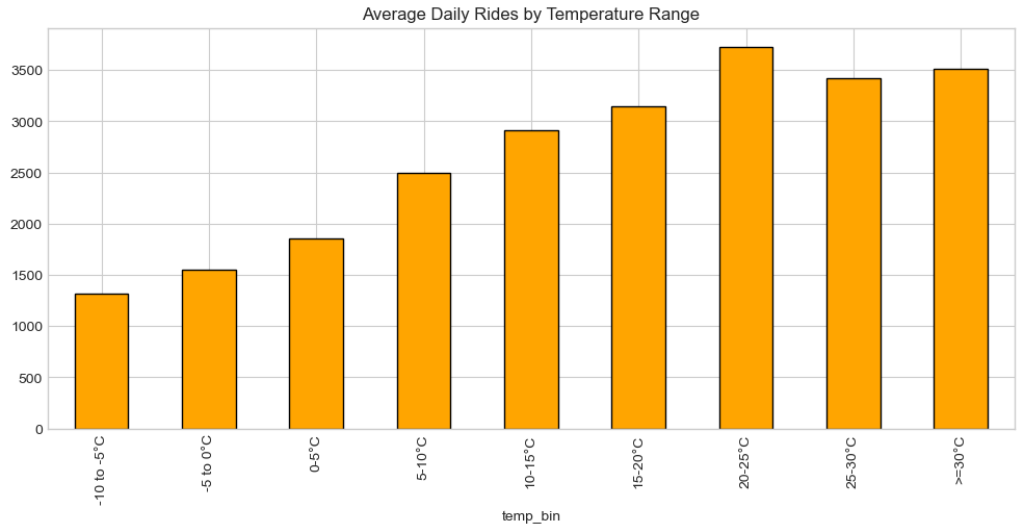

Figure 5: Rides by Temperature

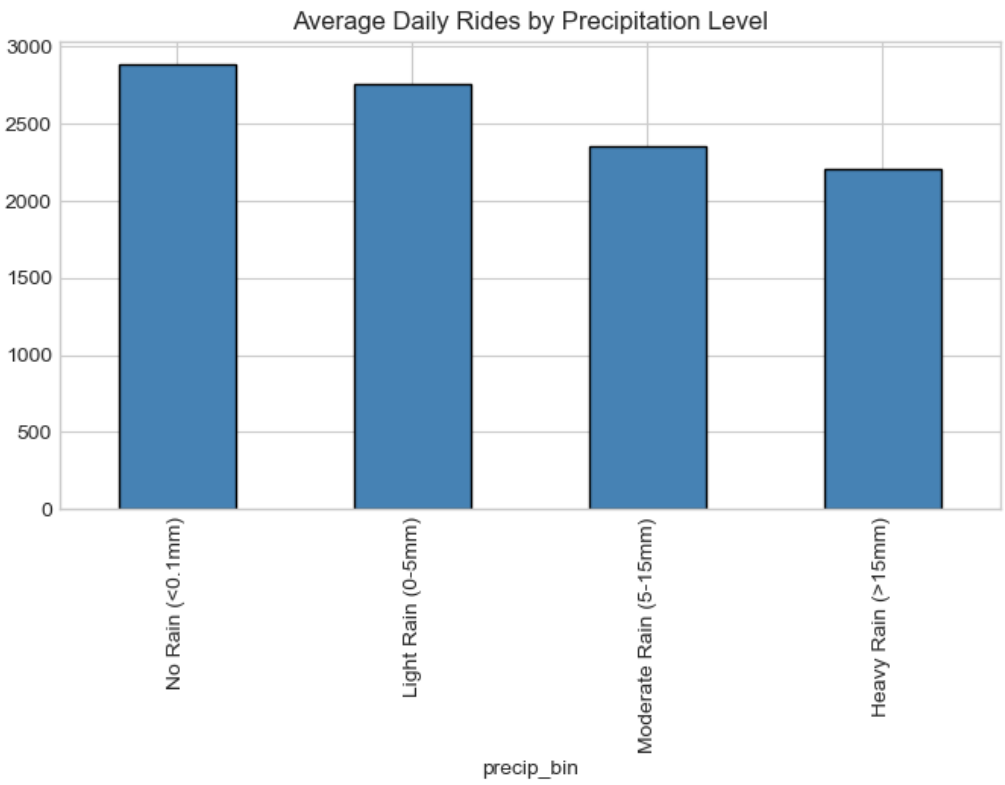

Figure 6: Rides by Precipitation

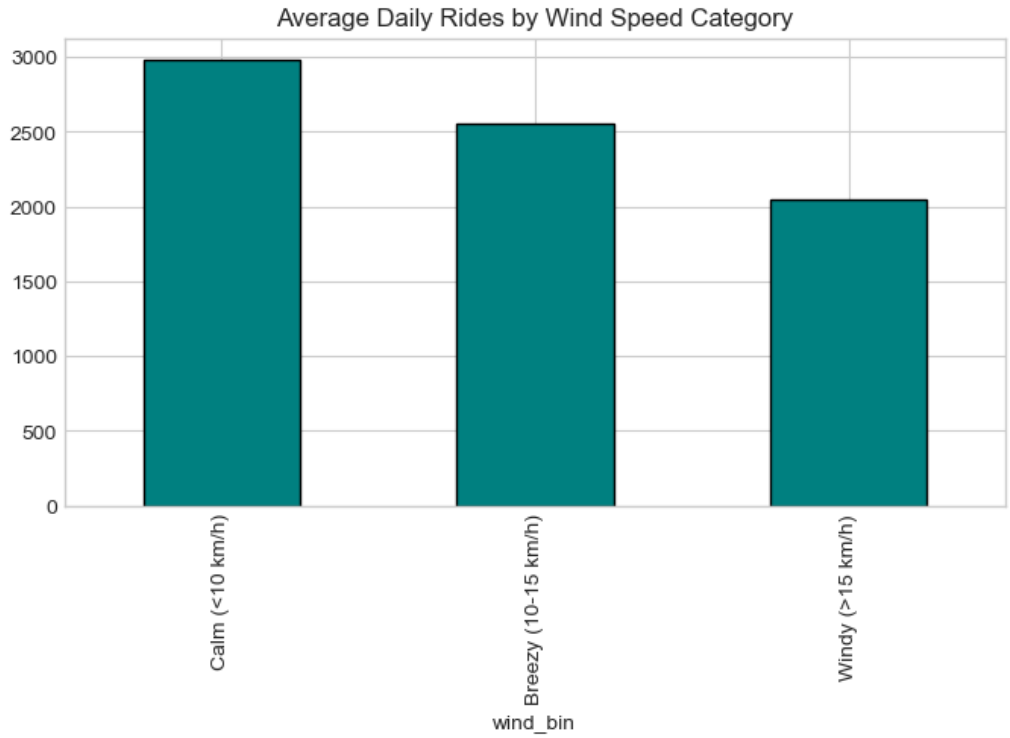

Figure 7: Rides by Wind Speed

4 Hourly Commute Pattern Analysis

Then we further investigate the data by looking at the hourly bike rides. We first compare the hourly distribution of trips for weekdays and weekends. In Figure 8, weekdays show a bimodal distribution with clear peaks during the morning (8:00 AM) and evening (5:00 PM - 6:00 PM) rush hours, while weekends do not have these commute spikes. Instead, they show a single, smooth peak in the afternoon, which aligns with leisure and social activities.

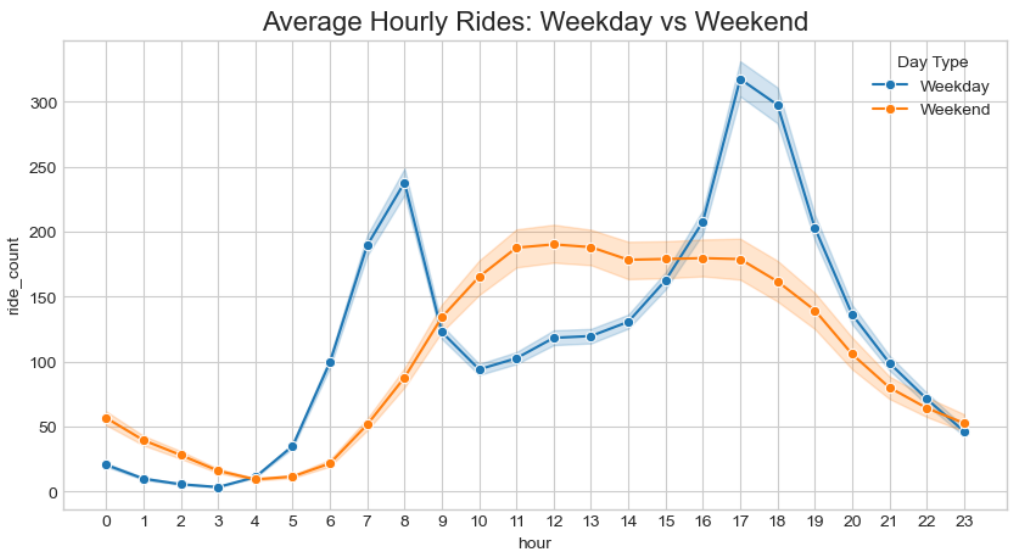

Figure 8: Average Hourly Rides: Weekday vs Weekend

4.1 Impact of Rain

Rain changes the typical hourly patterns. Figure 9 shows that even light or moderate rain reduces ride counts across all hours, but the peek in weekday rush hours still holds. Figure 10 further confirms this, showing that total ride volumes for both weekdays and weekends decrease significantly when it rains. Figure 11 shows the hourly distribution during heavy rain (around or above 7 mm/h). Notably, there is a significant drop in weekday morning rush hours. Also, it's worth noticing that a small peak appears in weekend rides, which are less effected by the heavy rain.

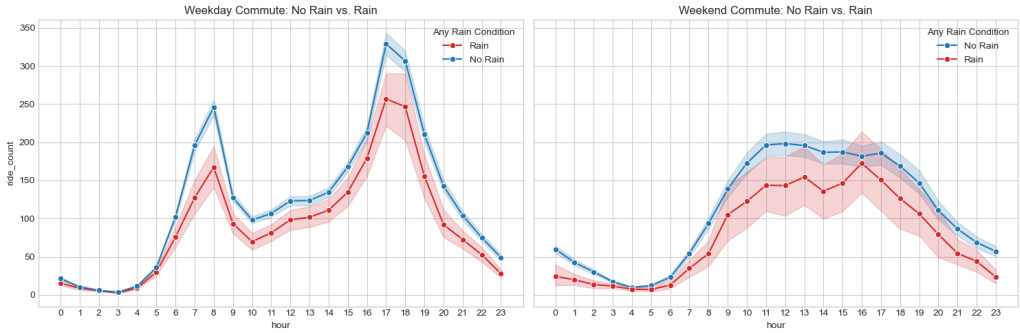

Figure 9: Hourly Commute by Rain

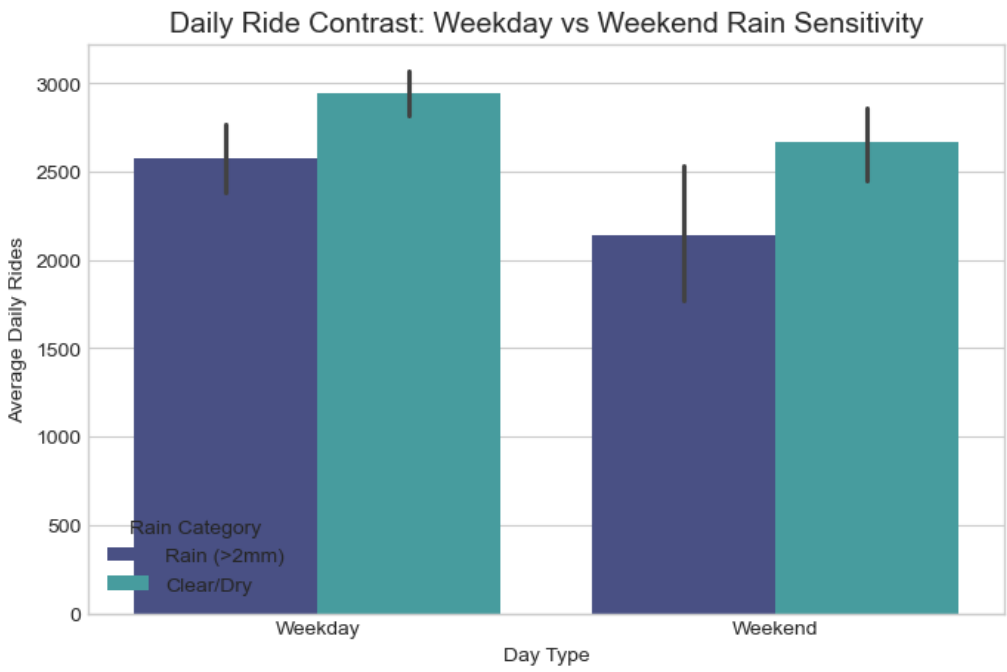

Figure 10: Daily Ride Rain Sensitivity

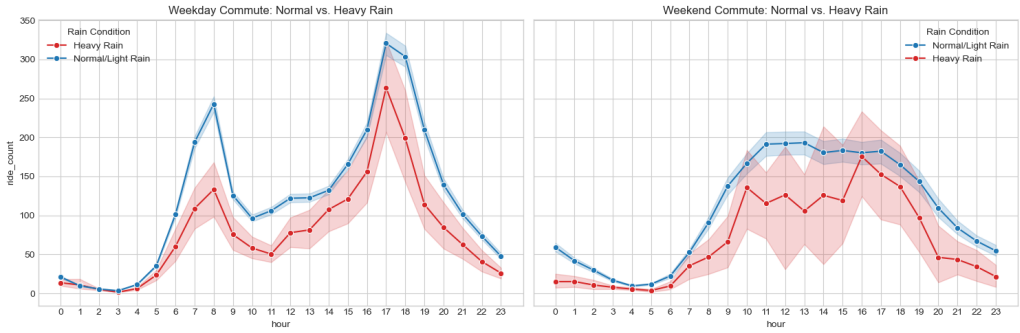

Figure 11: Hourly Commute by Heavy Rain

4.2 Impact of Wind

Wind also limits cycling to some extent. Figure 12 shows a slightly lower ridership at all hours on weekdays during extreme wind conditions (over 15km/h wind speed), while ridership on weekends is merely not affected. Figure 14 illustrates a sharp decline in overall daily volume due to wind. However, Figure 13 indicates that moderate wind (over 5km/h) only reduces the weekend ride counts and barely affect weekday patterns.

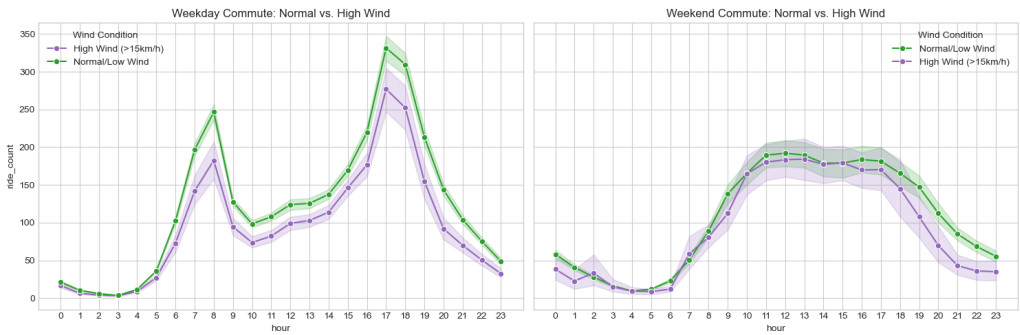

Figure 12: Hourly Commute by High Wind

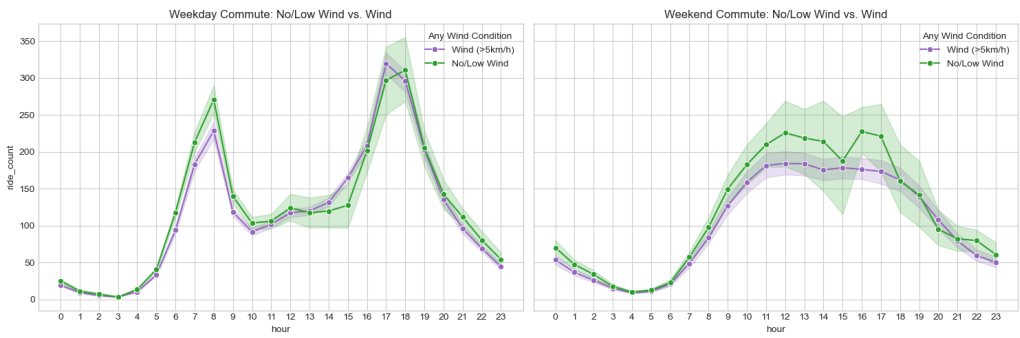

Figure 13: Hourly Commute by Moderate Wind

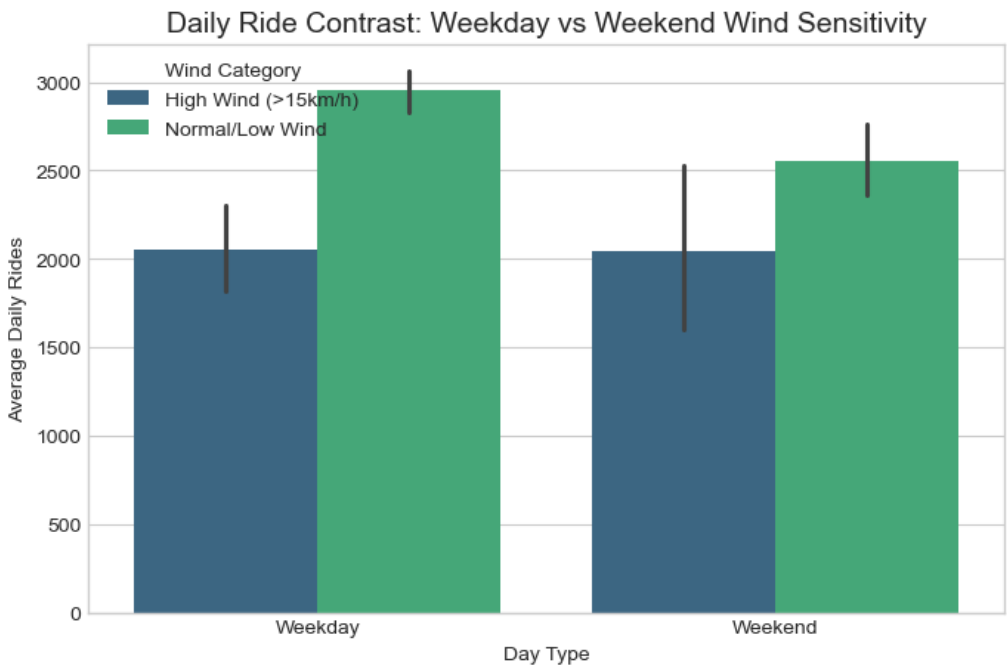

Figure 14: Daily Ride Wind Sensitivity

4.3 Impact of Temperature

Temperature influences ride volume similarly to rain and wind. Figure 15 shows hourly patterns grouped by different temperatures. Warm temperatures increase the height of morning and evening commute peaks. Freezing temperatures lower these peaks, as fewer people choose to ride in the cold. Figure 16 confirms the strong positive correlation between temperature and total daily rides, and the cold weather significantly cause the daily rides to drop.

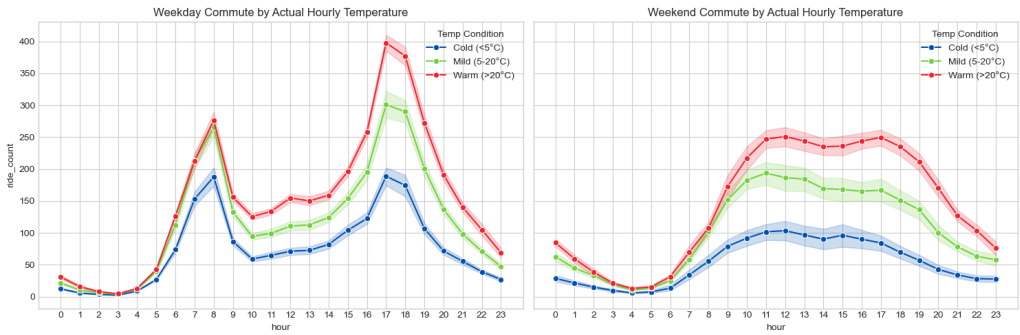

Figure 15: Hourly Commute by Temperature

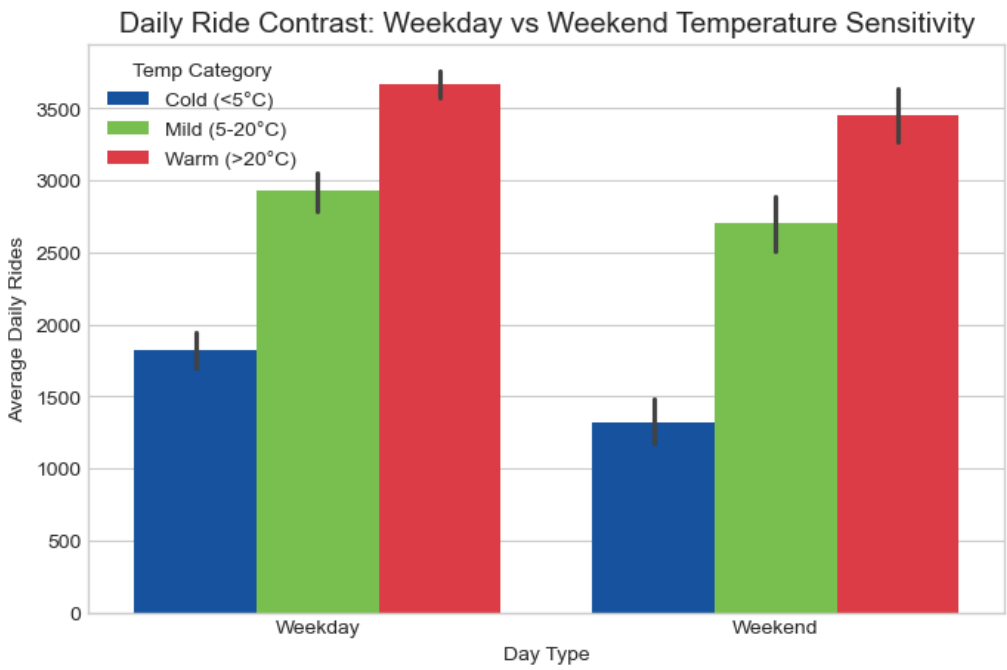

Figure 16: Daily Ride Temperature Sensitivity

4.4 User Types

We also separated the ridership by membership status to examine different user behaviors. Figure 17 shows that annual members account for most of the weekday rush hour peaks, confirming they use the bikes for daily commuting. Casual usage are much lower and more evenly distributed, since causal users only use the system for leisure.

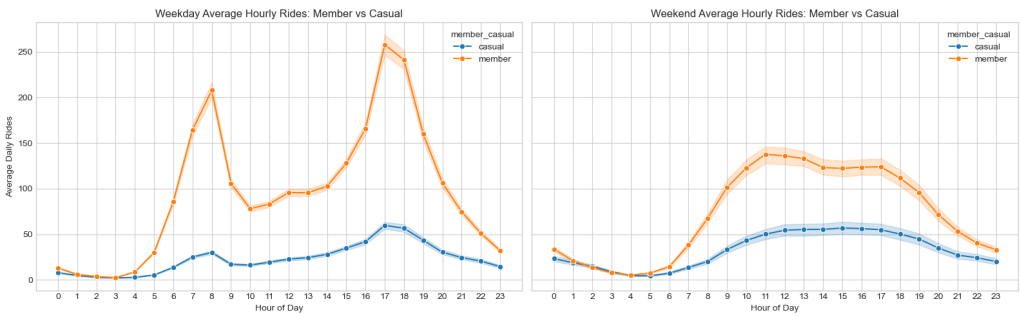

Figure 17: Average Hourly Rides by User Type

5 Rideable Type Preference

Lastly, we investigated whether electric bikes (e-bikes) change how users respond to weather. Figure 18 shows that the proportion of e-bike usage increases with higher wind speeds. This suggests riders use motor assistance to handle headwinds. On the other hand, Figure 19 and Figure 20 shows that the proportion of e-bike rides barely changes during hot or rainy weather.

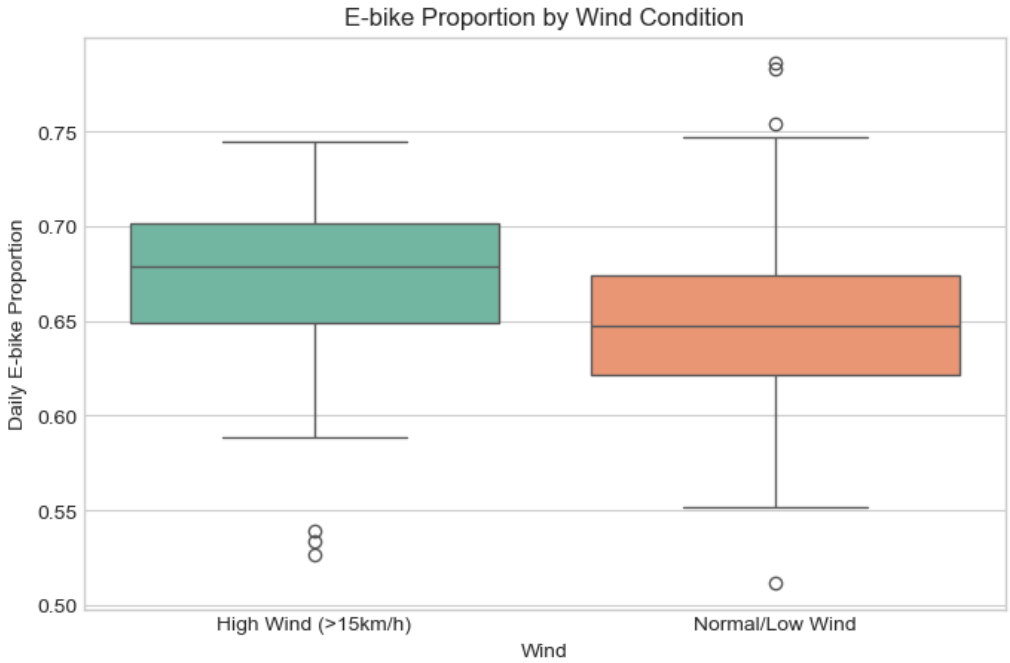

Figure 18: Preference by Wind

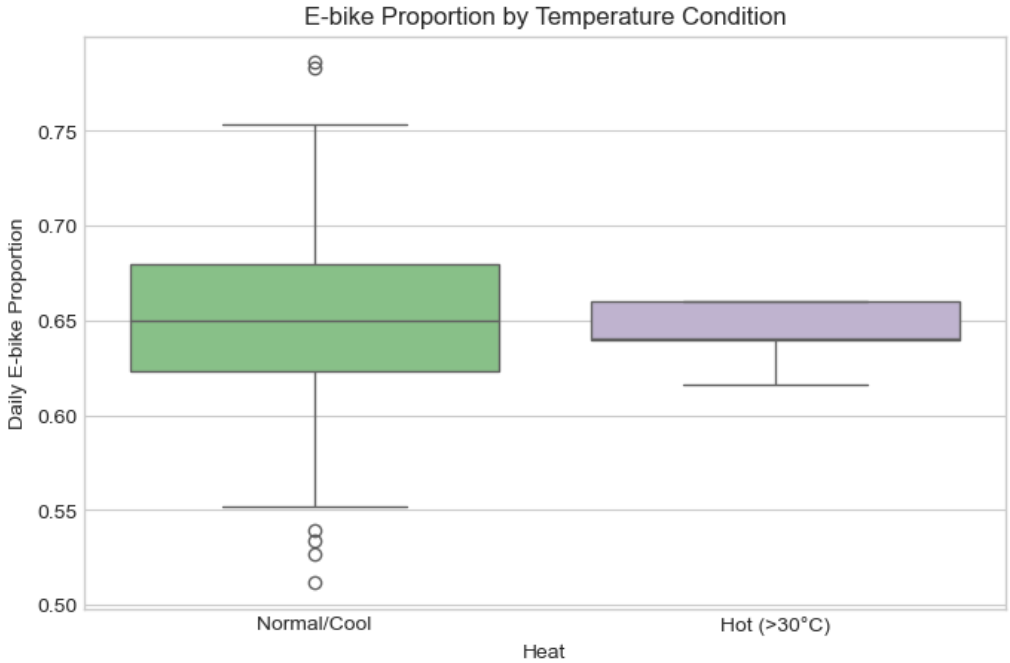

Figure 19: Preference by Temperature

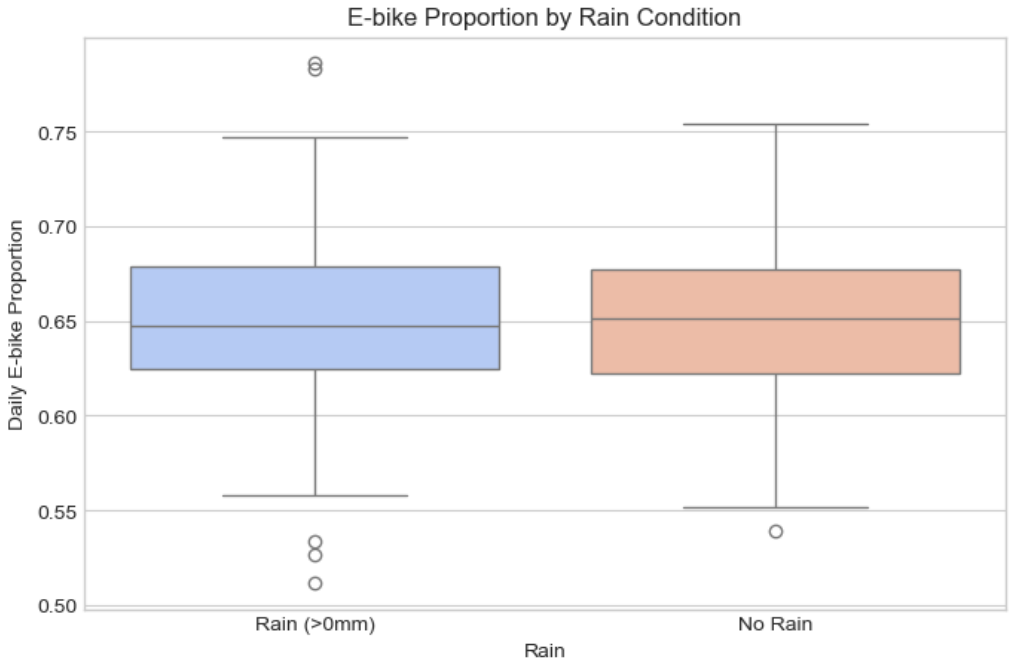

Figure 20: Preference by Rain

6 Conclusion

This analysis confirms that weather directly controls Citi Bike usage in Jersey City. Temperature is the main factor driving ride volume. Ridership increases steadily as temperatures rise, though it dips slightly on days hotter than 25 degrees Celsius. The system primarily serves commuters, which is shown by the clear morning and evening rush hour peaks on weekdays dominated by annual members.

Rain and wind clearly limit cycling, but they affect travel patterns differently. Rain causes the most significant drops in total rides. Light rain lowers overall volume while keeping the daily commute cycles intact, but heavy rain breaks the typical weekday morning rush hour. Wind reduces ridership by making riding physically harder. Interestingly, moderate wind mostly discourages weekend leisure riders, while extreme wind is required to noticeably reduce weekday commutes. This indicates that commuters are less likely to abandon their trips due to bad weather compared to casual riders.

Finally, the data shows a specific utility for electric bikes during adverse weather. Users actively choose e-bikes when facing strong winds, relying on the motor to offset the extra physical effort. However, the proportion of e-bike rides stays mostly unchanged during hot or rainy weather. This suggests that riders mainly use electric assistance to push through wind resistance, rather than to escape high temperatures or stay dry.